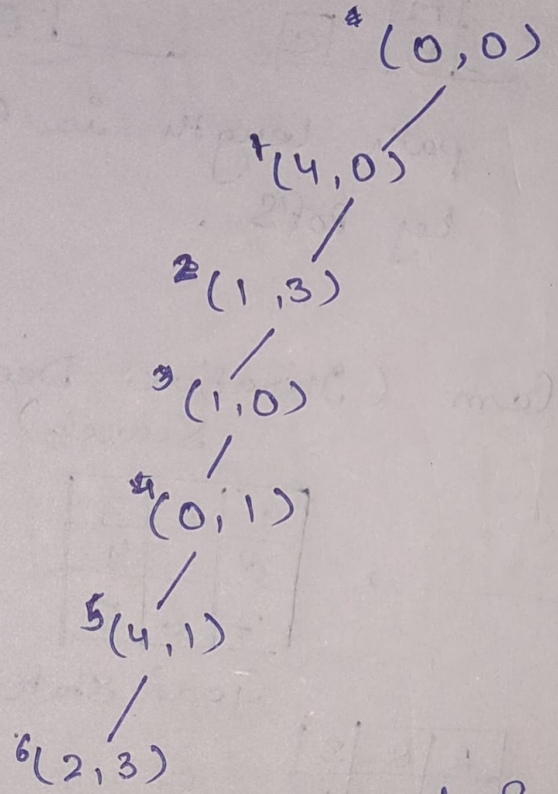
FOAI - Assignment 3

Q.1 Water Jug Problem (DFS)

we can represent the state as (x, y)
where, x ⟶ 4 gallon jug
y ⟶ 3 gallon jug.

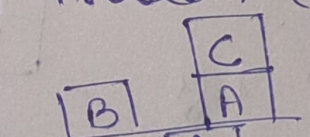
- Initial State : (0,0)
- Goal State : (2,y)



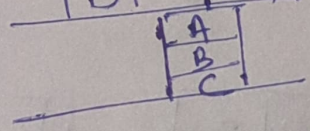
It took 6 nodes using DFS.

Q.2 Block World Problem (BFS)

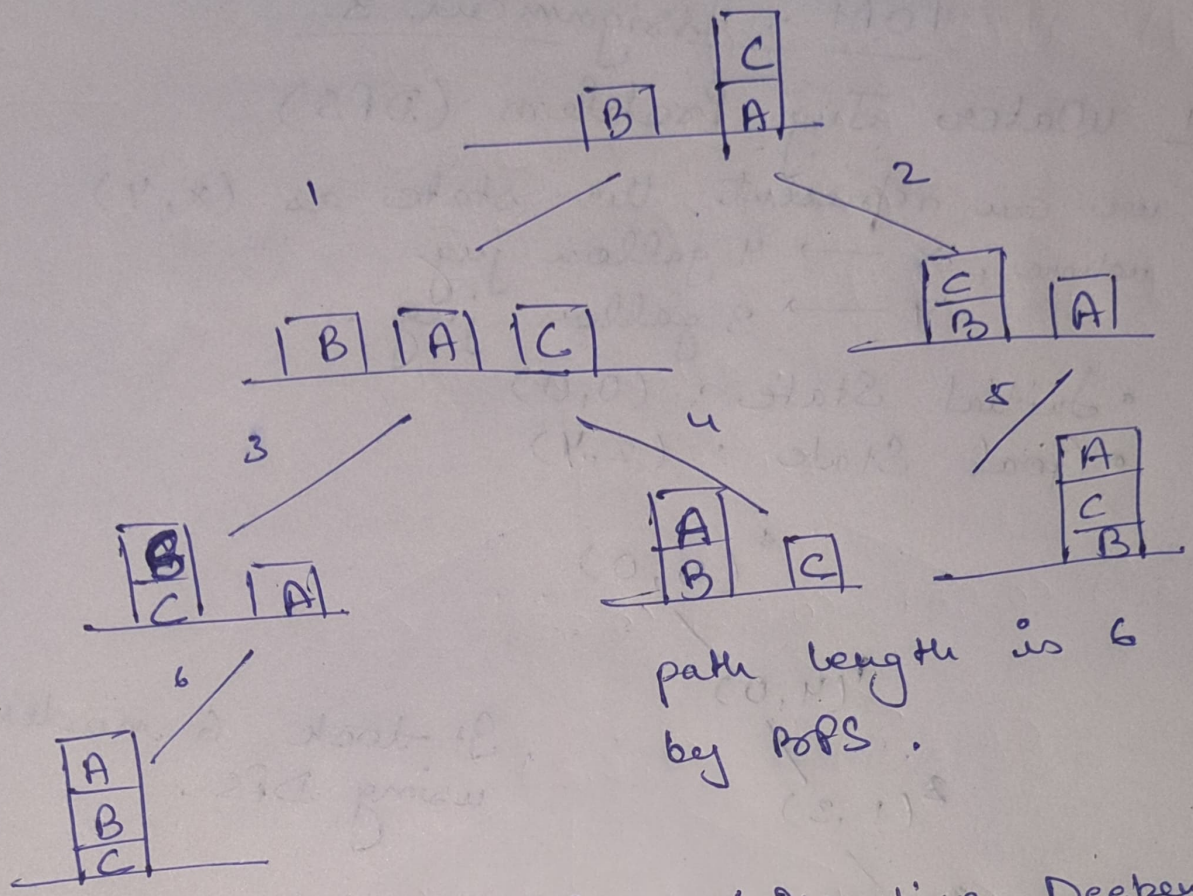
Initial State :



Final State :



using allowed stack (x, y), lift (x), putg(x) and unstack (x, y)



path length is 6 by BFS.

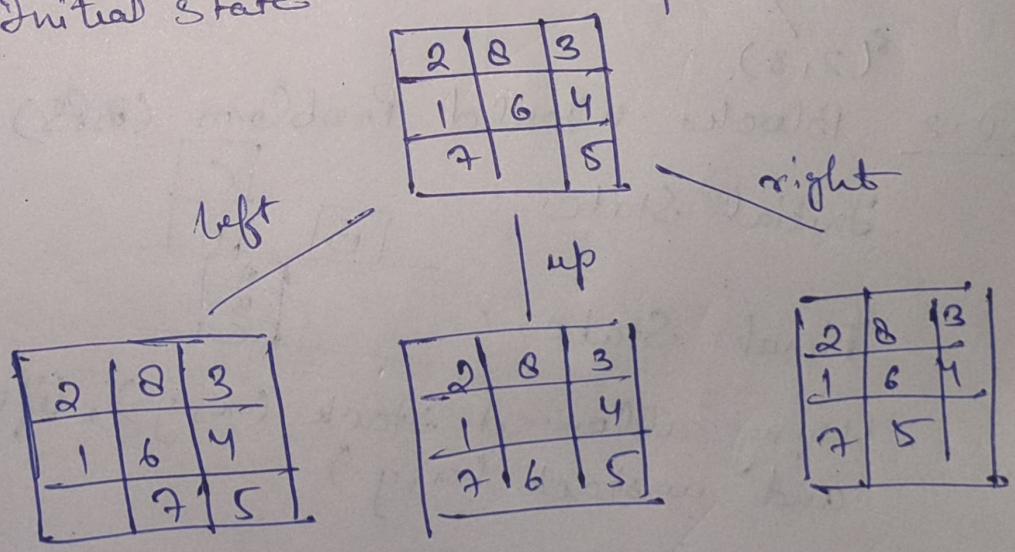
Q.3 8-Puzzle Problem (Iterative Deepening Search)

2	8	3
1	6	4
7		5

Initial state

1	2	3
8		4
7	6	5

Goal State



2	8	3
1		4
7	6	5

left

2	8	3
	1	4
7	6	5

up

2		3
1	8	4
7	6	5

right

2	8	3
1	4	
7	6	5

left

	2	3
1	8	4
7	6	5

right

2	3	
1	8	4
7	6	5

down

1	2	3
	8	4
7	6	5

down

1	2	3
7	8	4
	6	5

right

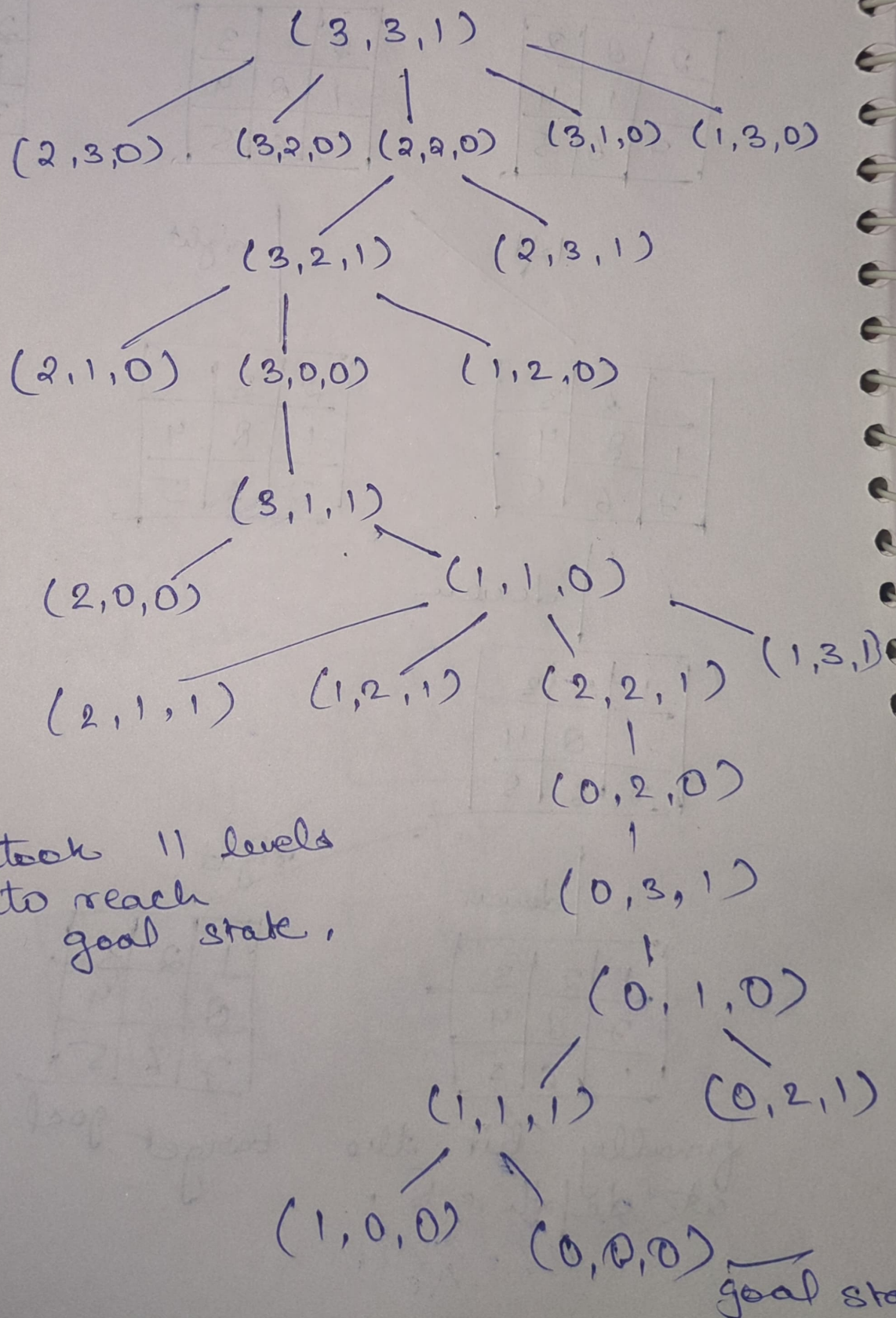
1	2	3
8		4
7	6	5

finally hit the target goal state.
at depth = 5.

Q.4 Cannibals and Missionaries (BFS)

- Initial State : (3, 3, 1)
- Goal State : (0, 0, 0)

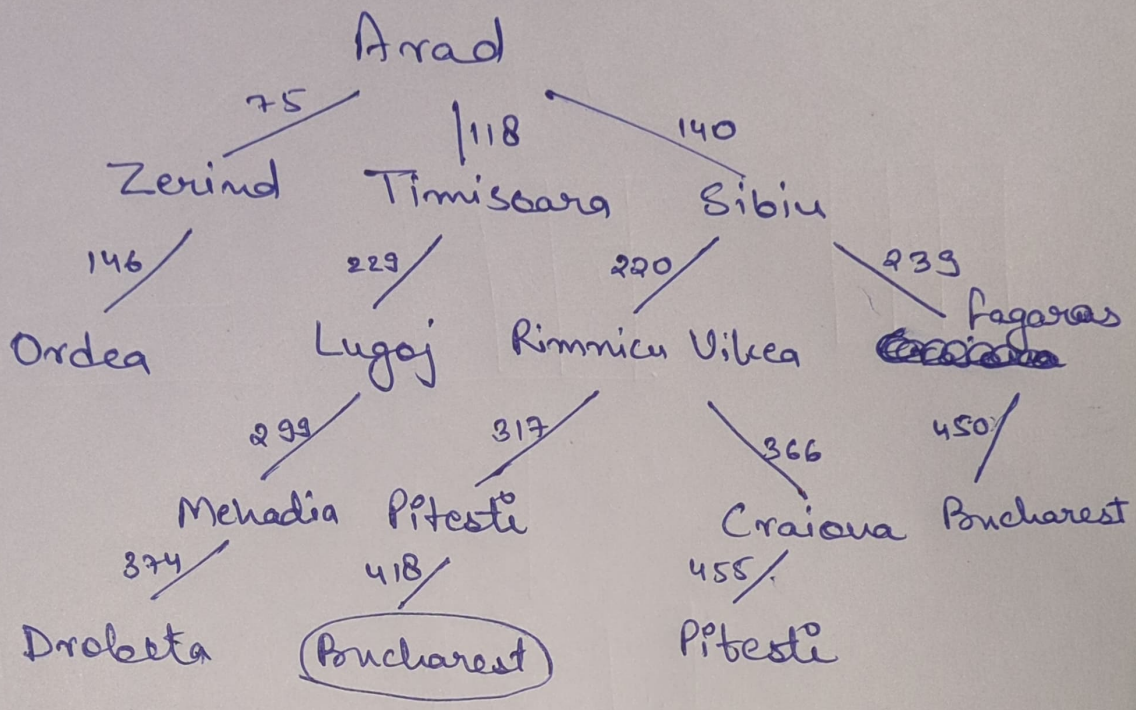
rule → • Cannibals cannot outnumber missionaries.



It took 11 levels to reach goal state.

Q.5 Map of Romania (UCS)

- Initial State : Arad (Cost = 0)
- Final State : Bucharest.



Final Optimal Path

→ Arad → Sibiu → Rimnicu Vilcea.
→ Pitesti → Bucharest

Total Cost : 418

UCS - expanded 12 nodes before sucessfully popping at Bucharest.